

OBJECTIVE

I am a recent graduate of the University of Cambridge with a MPhil in Data Intensive Science and a Senior Data Science Engineer at DraftKings. Outside of work, I was winner and runner-up in two hackathons and build hierarchical Bayesian models for horse racing.

SKILLS

Programming & Tools: Python, C++, SQL, Git, Docker, Jupyter, Pandas, NumPy, SciPy, Numba, scikit-learn, PyMC

PROFESSIONAL EXPERIENCE

DraftKings, London

Dec 2025 – Present

Senior Data Science Engineer – Trading Intelligence, Sports Data Science

- **Demand & Elasticity Modelling:** build demand and elasticity models from betting behaviour to estimate how odds changes impact betting volume.
- **Pricing & GGR Optimisation:** translate price movements into expected value and expected gross gaming revenue outcomes to support trading decisions.

Kantar, London

Jul 2023 – Nov 2025

Data Scientist – Forecasting & Analytics

- **End-to-End Panel Health Pipeline:** designed and maintained a Databricks-Snowflake pipeline ingesting 50M+ records weekly, automating data transformation, and creating dashboard metrics to monitor panel health.
- **Software Engineering & Architecture:** refactored and modularised legacy Python and SQL codebases, building reusable libraries for use across multiple services. Constructed bash pipelines for linux server interactions.

Amazon, East London

Jul 2022 – Oct 2022

Operations & Logistics Associate

- **Data Pipeline:** developed a scraping workflow on internal dashboards in order to categorise the root cause of misrouting issues.
- **Data Integration & Compliance Checks:** integrated compliance checks for delivery routes to pattern wider trends.
- **Cross-Functional Analysis & Recognition:** authored and presented a detailed whitepaper on pipeline methodology and impact, earning an Operational Excellence award.

EDUCATION

University of Cambridge

Oct 2024 – August 2025

MPhil in Data Intensive Science (Distinction)

- **MPhil Thesis (Jul 2025): T3Net - Joint PDF & SMEFT Inference:** Created neural-network parameterisation to extract non-singlet PDFs and SMEFT Wilson coefficients. Applied replica-based Monte Carlo training with Lagrange penalties to propagate uncertainties and test against ongoing global QCD analysis.
- **Capital Markets Fundamentals:** completion of Cambridge Summer School in Finance, covering fixed income, derivatives pricing and market risk management.

University College London

Oct 2019 – Jul 2023

Physics MSci (Second Class Honours, Upper Division)

- **MSci Thesis (Apr 2023): Quantum Track Reconstruction:** Approached LHC track reconstruction as a graph classification problem with hybrid quantum-classical GNNs. Integrated parameterised quantum circuits and tested “entangling dropout” to explore classical analogues of quantum ML
- **Achievements:** Awarded first prize for a cohort-wide statistical analysis of maritime scrubber emissions

PERSONAL PROJECTS

Bayesian Horse Racing Model

Present

- Scraped and engineered structured datasets from the Horse Racing Association.
- Extracted historical results, horse/jockey/trainer metadata, and environmental conditions.
- Designed and implemented a hierarchical Bayesian logistic regression model in PyMC to estimate win probabilities.

Custom Parachain for Insurance Workflows (Hackathon Winner)

2023

- Conceptualised and prototyped a Polkadot parachain using zero-knowledge proof contracts to combine NHS records, employment history and smart-car telematics for privacy-preserving premium calculation.
- Authored test scenarios and demo scripts illustrating secure ZKP-based information flows and GDPR-compliant data exchanges.

Dynamic Art NFT Platform on Tezos (Hackathon Runner Up)

2022

- Prototype NFT ecosystem where artworks adapt their metadata in response to simulated audience engagement metrics.
- Demonstrated the concept of merging blockchain interactivity with cultural experiences in a post-pandemic context.